

Quiz: Sorting

Total Score: $/2^2$

Printed Name:

Quiz rules:

1. You MAY use any printed or handwritten notes.
2. You MAY NOT use a computer or any other electronic device.

Problem 1. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def foo(xs):
4     def go(i, acc):
5         if i == len(xs):
6             return acc
7         if xs[i] % 2 == 0:
8             return go(i+1, acc - xs[i])
9         return go(i+1, acc + xs[i])
10    return go(0, 0)
11 try:
12     print('foo([1, 2, 3, 4, 5])=',foo([1, 2, 3, 4, 5]))
13 except RuntimeError:
14     print('StackOverflow')
15 EOF
16 $ python3 foo.py
```

Fraction of LLMs with correct answer: $14 / 16 = 0.88$

Problem 2. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 xs = ['kiwi', 'fig', 'cherry', 'banana', 'date', 'apple']
4 zs = list(sorted(xs, key=lambda w: len(w)))
5 print('zs=', zs)
6 EOF
7 $ python3 foo.py
```

Fraction of LLMs with correct answer: $14 / 16 = 0.88$

Problem 3. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def merge_sorted(xs):
4     if len(xs) <= 1:
5         return xs
6     mid = len(xs) - 1
7     return merged(merge_sorted(xs[:mid]), merge_sorted(xs[mid:]))
8
9 def merged(xs, ys):
10     out = []
11     i = 0
12     j = 0
13     while i < len(xs) and j < len(ys):
14         if xs[i] <= ys[j]:
15             out.append(xs[i])
16             i += 1
17         else:
18             out.append(ys[j])
19             j += 1
20     out += xs[i:]
21     out += ys[j:]
22     return out
23
24 xs = [3, 7, 1, 8, 5, 10, 2, 9, 4, 6]
25 try:
26     zs = merge_sorted(xs)
27     print('zs=', zs)
28 except RuntimeError:
29     print('StackOverflow')
30 EOF
31 $ python3 foo.py
```

Fraction of LLMs with correct answer: 13 / 16 = 0.81

Problem 4. Write the output of the final command in the following shell script.

```
1 $ cd; rm -rf quiz; mkdir quiz; cd quiz
2 $ cat > foo.py <<EOF
3 def quick_sorted(xs):
4     if len(xs) <= 1:
5         return xs
6     mid = len(xs) // 2
7     pivot = xs[mid]
8     smaller = [x for x in xs if x <= pivot]
9     larger = [x for x in xs if x >= pivot]
10    return quick_sorted(smaller) + quick_sorted(larger)
11
12 xs = [3, 7, 1, 8, 5, 10, 2, 9, 4, 6]
13 try:
14     zs = quick_sorted(xs)
15     print('zs=', zs)
16 except RuntimeError:
17     print('StackOverflow')
18 EOF
19 $ python3 foo.py
```

Fraction of LLMs with correct answer: 14 / 16 = 0.88

LLM Model Performance

